

Enabling High-Order QAM OFDM for Satellite Feeder Links via PGIR-Integrated Companding

Li Cho

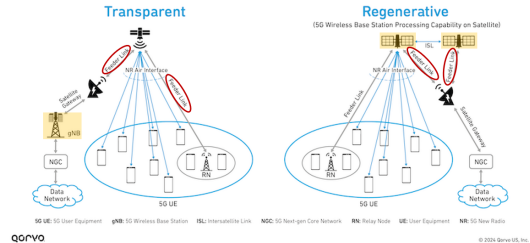
Department of Electrical Engineering
Tatung University, Taipei, Taiwan
lcho@gm.ttu.edu.tw

Manuscript submitted to IEEE Transactions on Communications
2026

- ➊ **Background and Motivation:** feeder-link bottleneck, PAPR vs. HPA, companding lineage
- ➋ **System Model:** OFDM transceiver, SSPA / TWTA, shadowed-Rician channel
- ➌ **Proposed Compander + PGIR:** C^1 design, receiver pipeline, convergence
- ➍ **Saturation-Ratio Optimization:** NMSE decomposition, three regimes, BER proxy, CSI gate
- ➎ **Numerical Results:** PAPR, spectrum, convergence, amplifiers, modulation scaling
- ➏ **Conclusion**
- ➐ **Appendix:** C^1 tail rate and uniqueness of g , noise-unbiased AGC

6G NTN and the Feeder-Link Bottleneck

- ▶ 6G → fully integrated **non-terrestrial network**: maritime, aerial, remote coverage
- ▶ Feeder links = backbone: terrestrial gateways ↔ LEO mega-constellations, HTS
- ▶ Tbps-class aggregate capacity ⇒ feeder link = **throughput bottleneck** of the whole NTN
- ▶ Trend: Q / V bands, wider bandwidth, harsher impairments
- ▶ Spectral-efficiency KPIs ⇒ **OFDM + high-order QAM indispensable**



Satellite Payload.

Scenario

Feeder-link **downlink**: on-board HPA near saturation at the satellite, compute-rich gateway at the ground.

[1] W. Saad *et al.*, "A vision of 6G wireless systems," *IEEE Netw.*, 2020. [6] R. Deng *et al.*, "Ultra-dense LEO satellite constellations," *IEEE Trans. Wireless Commun.*, 2021. [7] I. Ahmad *et al.*, "Next-generation high throughput satellite systems," *IEEE Syst. J.*, 2021.

Figure source: www.qorvo.com

The PAPR–HPA Problem

- ▶ Scarce on-board DC power $\Rightarrow$ HPA near saturation for power-added efficiency
- ▶ Nonlinear **AM/AM + AM/PM**, dense QAM constellations: little error margin
- ▶ High OFDM PAPR $\Rightarrow$ peaks hit deep saturation
 - ▷ in-band distortion $\rightarrow$ **BER floors** no noise-limited receiver removes
 - ▷ spectral regrowth $\rightarrow$ OOB mask violations in congested orbital slots
- ▶ 3GPP: PAPR flagged in NTN downlink [14], [15], Rel-16 / Rel-19 studies, yet **no standardized PHY remedy** [18]

Need

PAPR suppression with no throughput loss, no power-efficiency loss.

[14] Huawei, HiSilicon, "Discussion on PAPR in NTN," 3GPP Tdoc R1-1907516, 2019. [15] Thales, "Considerations on PAPR requirements for NR NTN downlink," 3GPP Tdoc R1-2005311, 2020. [18] 3GPP, "NR; physical channels and modulation," TS 38.211 V19.2.0, 2025.

Companding: Three Decades, One Frontier

No side information, no bandwidth expansion, low complexity. Each generation relaxed the previous binding constraint:

- 1 **Logarithmic** (μ -law, A-law) [21]: dynamic-range compression, but average power rises $\rightarrow$ HPA budget tightens
- 2 **Nonlinear transforms** (EC, trapezoidal, piecewise-exp.) [24]: target densities at constant power, but small-signal expansion amplifies noise
- 3 **Piecewise linear (PLC)** [27]: low complexity, minimal distortion, but hard clipping = many-to-one $\rightarrow$ **irreducible BER floor** at high-order QAM
- 4 **GPLC + optimized / adaptive designs** [30]: one-to-one compression, variable power, up to 512-QAM
- 5 **State of the art [37]**: SDR-optimal transform is piecewise linear, best known PAPR-BER-OOB trade-off, *transmitter-only*

[21] X. Wang *et al.*, *IEEE Trans. Broadcast.*, 1999. [24] T. Jiang *et al.*, *IEEE Trans. Broadcast.*, 2005. [27] M. Hu *et al.*, *IEEE Trans. Broadcast.*, 2014. [30] M. Hu *et al.*, *IEEE Trans. Broadcast.*, 2020. [37] M. Hu *et al.*, "Near-optimal piecewise linear companding transform for PAPR reduction of OFDM systems," *IEEE Trans. Broadcast.*, 2025.

The Missed Lever: Co-Design with the Receiver

Constraint across the whole lineage

All schemes act **unilaterally at the transmitter**. Receiver = passive decompaning + equalization. No recourse after the compander.

- ▶ Receiver-side recovery exists, but always *independent* of the compander:
 - ▷ decision-aided cancellation [38]: tied to detection reliability
 - ▷ compressive sensing [40]: sparsity prior broken by companding
 - ▷ PA-inverting equalizer [41]: needs an explicit amplifier model
- ▶ **PGIR** [42]–[44]: rebuild discarded samples from known **null subcarriers**. No side info, no decisions, no prior, no PA model

This paper closes the loop

Compander **designed for PGIR**: it opens the classification margin reconstruction needs.

[38] H. Chen and A. M. Haimovich, *IEEE Commun. Lett.*, 2003. [40] E. B. Al-Safadi and T. Y. Al-Naffouri, *IEEE Trans. Signal Process.*, 2012. [41] J. He et al., *IEEE Trans. Commun.*, 2024. [42] R. W. Gerchberg, *Optica Acta*, 1974. [43] A. Papoulis, *IEEE Trans. Circuits Syst.*, 1975. [44] L. Cho and S.-W. Chang, *IEEE Signal Process. Lett.*, 2026.

Contributions

- ① **PGIR-oriented compander**: smooth tail, saturation ratio $\eta \in (0, 1]$, classification margin. C^1 continuity $\Rightarrow \mathcal{O}(1/f^3)$ OOB decay, closed form. Hard clipper of [37] = degenerate case $\eta = 1$
- ② **SNR-adaptive η optimization**: two-term NMSE decomposition $\Rightarrow$ interior optimum, **three SNR regimes**. Hard limiter optimal at low SNR
- ③ **NMSE as BER proxy**: BER strictly increases with NMSE, independent of modulation order $\Rightarrow \eta$ from cheap profiling, no BER Monte Carlo
- ④ **AM/PM immunity**: saturated samples discarded and rebuilt from the null-subcarrier subspace $\Rightarrow$ TWTA phase corruption removed, **no Saleh parameters needed**

[37] M. Hu *et al.*, "Near-optimal piecewise linear companding transform for PAPR reduction of OFDM systems," *IEEE Trans. Broadcast.*, 2025.

System Model: Transceiver and Signal

- ▶ N_{sc} active subcarriers, oversampling L , DFT size $N = LN_{\text{sc}}$, unit-power square M -QAM, $N - N_{\text{sc}}$ null subcarriers

$$\mathbf{x} = \mathbf{F}^H \mathbf{X}, \quad \text{PAPR} = \max_n |x_n|^2, \quad f_R(r) = 2re^{-r^2} \text{ (Rayleigh envelope)}$$

- ▶ Envelope companding before the HPA: $\tilde{x}_n = \mathcal{C}(|x_n|)e^{j\angle x_n}$

$$\mathbf{y} = h \mathcal{G}_{\text{pa}}(\tilde{\mathbf{x}}) + \mathbf{w}, \quad \mathbf{w} \sim \mathcal{CN}(\mathbf{0}, \sigma_w^2 \mathbf{I}), \quad \sigma_w^2 = L/\gamma_s$$

- ▶ $\mathcal{G}_{\text{pa}}$: PA nonlinearity, h : flat fading, $\gamma_s \triangleq E_s/N_0$: receive symbol SNR
- ▶ Unit-power signal spread over the full oversampled band $\Rightarrow$ noise scales with L

[45] H. Ochiai and H. Imai, "On the distribution of the peak-to-average power ratio in OFDM signals," *IEEE Trans. Commun.*, 2001.

Power Amplifier Models

Rapp SSPA: AM/AM only

$$A_{\text{out}}(r) = \frac{r}{[1 + (r/A_{\text{sat}})^{2p_{\text{rapp}}}]^{1/(2p_{\text{rapp}})}}$$

- ▶ smooth, mild distortion on all samples
- ▶ $A_{\text{sat}} \geq R \Rightarrow$ near-linear over $[0, R]$

Saleh TWTA: AM/AM + AM/PM

$$A_{\text{out}}(r) = \frac{\alpha_S r}{1 + \beta_S r^2}, \quad \Phi_{\text{out}}(r) = \frac{\alpha_\Phi r^2}{1 + \beta_\Phi r^2}$$

- ▶ phase rotation kills dense QAM
- ▶ gain folds back at $r^* = 1/\sqrt{\beta_S}$: many-to-one
- ▶ $(\alpha_\Phi, \beta_\Phi)$ unavailable in practice

Key observation

Compression + phase rotation hit exactly the samples the receiver **discards and reconstructs** $\Rightarrow$ TWTA handled with zero PA knowledge.

[46] C. Rapp, *Proc. 2nd Eur. Conf. Satell. Commun.*, 1991.
Commun., 1981.

[47] A. A. M. Saleh, "Frequency-independent and frequency-dependent nonlinear models of TWT amplifiers," *IEEE Trans.*

Shadowed-Rician Channel (SR-AS)

$$h = \sqrt{\frac{K_{\text{ric}}}{K_{\text{ric}}+1}} h_{\text{LOS}} + \sqrt{\frac{1}{K_{\text{ric}}+1}} h_{\text{sca}}, \quad |h_{\text{LOS}}|^2 \sim \text{Gamma}(m, 1/m), \quad h_{\text{sca}} \sim \mathcal{CN}(0, 1)$$

- ▶ Slant-path shadowing (terrain, foliage), severity set by m
- ▶ Coherence bandwidth $\gg$ subcarrier spacing $\Rightarrow$ frequency flat, h constant per symbol
- ▶ Perfect CSI, Doppler pre-compensation, $\mathbb{E}[|h|^2] = 1$
- ▶ Average shadowing: $(b_0, m, \Omega) = (0.126, 10.1, 0.835)$, $K_{\text{ric}} = 3.31$ (5.2 dB)

Why fading matters

Equalization rescales noise by $|h|^{-2}$. The **deep-fade tail** decides where reconstruction helps and where it hurts $\rightarrow$ CSI gate (Sec. IV).

[48] A. Abdi, W. C. Lau, M.-S. Alouini, and M. Kaveh, "A new simple model for land mobile satellite channels: First- and second-order statistics," *IEEE Trans. Wireless Commun.*, 2003.

Proposed C^1 Compaander

$$\mathcal{C}(r) = \begin{cases} gr, & 0 \leq r \leq A_{\text{th}}, \\ \eta R + (1 - \eta)R[1 - e^{-\beta(r^2 - A_{\text{th}}^2)}], & r > A_{\text{th}} \end{cases}$$

- R : target crest factor $\Rightarrow$ PAPR capped at R^2
- $\eta \in (0, 1]$: **saturation ratio**, the design freedom
- linear branch up to $A_{\text{th}} = \eta R/g$, smooth exponential tail toward R
- C^1 transition vs. the hard limiter's derivative kink
- Saleh fold-back region: harmless, those samples are **never inverted**

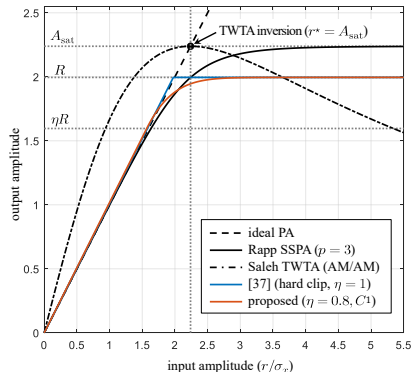


Fig. 1: AM/AM, $R = 6$ dB.

[37] M. Hu *et al.*, *IEEE Trans. Broadcast.*, 2025 (hard clip, $\eta = 1$). [46] C. Rapp, 1991. [47] A. A. M. Saleh, *IEEE Trans. Commun.*, 1981.

Closed-Form Parameters

Slope g : energy conservation $\mathbb{E}[|\mathcal{C}(r)|^2] = 1$ over the Rayleigh envelope

$$g^2 [1 - e^{-A_{\text{th}}^2} (1 + A_{\text{th}}^2)] + R^2 e^{-A_{\text{th}}^2} \left[1 - \frac{2(1 - \eta)}{\beta + 1} + \frac{(1 - \eta)^2}{2\beta + 1} \right] = 1$$

Tail rate β : C^1 matching at A_{th} (Appendix A)

$$\beta = \frac{g^2}{2\eta(1 - \eta)R^2}, \quad \mathcal{U} = \Pr(r > A_{\text{th}}) = e^{-A_{\text{th}}^2} \text{ (unknown ratio)}$$

- ▶ One implicit equation in g per (R, η) , **unique root for every** $\eta \in (0, 1)$
- ▶ Tail rate in **closed form**, vs. numerical search in [37]
- ▶ $\eta = 1$: saturating branch vanishes $\rightarrow$ hard limiter of [37]

[37] M. Hu *et al.*, "Near-optimal piecewise linear companding transform for PAPR reduction of OFDM systems," *IEEE Trans. Broadcast.*, 2025.

Receiver Pipeline: Discard and Reconstruct

$$\mathcal{C}^{-1}(v) = \begin{cases} v/g, & 0 \leq v < \eta R \text{ (known)}, \\ v, & v \geq \eta R \text{ (unknown)} \end{cases}$$

- ▶ equalize, AGC referral, threshold at ηR
- ▶ **no PA knowledge anywhere**
- ▶ (η, R) semi-static config $\Rightarrow$ **no per-symbol signaling**
- ▶ tail passed through, never inverted
- ▶ $g > 1 \Rightarrow$ known samples' noise attenuated by $1/g^2$: cleaner basis for PGIR
- ▶ $\eta = 1$: PGIR bypass, transceiver $\equiv$ [37]

Algorithm 1 Receiver pipeline (per OFDM symbol).

Require: $\mathbf{y}$, CSI h , AGC scalar G_{agc} (Appendix B);
 $\eta, R, g, i_{\text{max}}$

Ensure: active-subcarrier estimate $\hat{\mathbf{s}}$

- 1: $\mathbf{z} \leftarrow G_{\text{agc}}(\mathbf{y}/h)$ ▷ equalize + AGC referral
 - 2: $\mathcal{S} \leftarrow \{n : |z_n| \geq \eta R\}$ ▷ unknown (saturated) set
 - 3: $\hat{x}_n \leftarrow z_n/g, n \notin \mathcal{S}$ ▷ known: decompand (12)
 - 4: $\hat{x}_n \leftarrow z_n, n \in \mathcal{S}$ ▷ unknown: pass-through init
 - 5: **if** $\eta < 1$ **then** ▷ PGIR; $\eta = 1$: bypass
 - 6: **for** $i = 1, \dots, i_{\text{max}}$ **do**
 - 7: $\mathbf{X} \leftarrow \mathbf{F}\hat{\mathbf{x}}$; null subcarriers $\leftarrow 0$
 - 8: $\hat{\mathbf{x}} \leftarrow \mathbf{F}^H \mathbf{X}$; $\hat{x}_n \leftarrow z_n/g, n \notin \mathcal{S}$
 - 9: **end for**
 - 10: **end if**
 - 11: $\hat{\mathbf{s}} \leftarrow (\mathbf{F}\hat{\mathbf{x}})|_{\text{active}}$ ▷ DFT demodulation
-

[44] L. Cho and S.-W. Chang, "On the performance of discrete Papoulis–Gerchberg type iterative reconstruction," *IEEE Signal Process. Lett.*, 2026.

Smoothness–OOB Trade-off

- ▶ OOB decay set by smoothness at A_{th} :
 - ▷ C^0 kink $\rightarrow$ field amplitude $\mathcal{O}(1/f^2)$
 - ▷ C^1 condition $\rightarrow \mathcal{O}(1/f^3)$ for every $\eta \in (0, 1)$
- ▶ $\eta \rightarrow 1$: $\beta \rightarrow \infty$, collapse to C^0 hard limiter $\Rightarrow \eta = 1 =$ **degenerate endpoint** of the design space $(0, 1]$

Two roles of the smooth tail (neither touches reconstruction accuracy)

- 1 $\mathcal{O}(1/f^3)$ spectral decay
- 2 $\eta < 1$ opens a **classification margin**: peaks live in $[\eta R, R]$, away from the linear region $\Rightarrow$ robust threshold decisions under noise

Smaller η : wider margin but larger $\mathcal{U} \Rightarrow$ tension resolved in Sec. IV.

PGIR Convergence

- ▶ alternate: known samples fixed $\leftrightarrow$ energy on active subcarriers
- ▶ unknown-set error operator

$$\mathbf{S} = \mathbf{I}'_D \mathbf{F}^H \mathbf{I}'_T \mathbf{F}, \quad \rho(\mathbf{S}) < 1 \Rightarrow \text{contraction}$$

- ▶ necessary: $\mathcal{U} < \mathcal{K} = (L - 1)/L$, discarded samples outnumbered by null constraints
- ▶ nulls = oversampling headroom, **zero data-rate cost**
- ▶ $L \geq 4$: $\mathcal{K} \geq 3/4$, wide margin in practice
- ▶ saturated set = contiguous excursions $\Rightarrow \rho(\mathbf{S}) \uparrow$ as $\mathcal{U}$ grows (smaller R), larger L relaxes

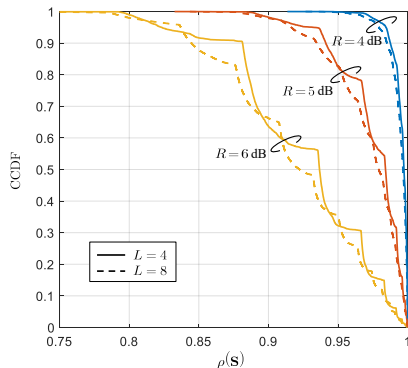


Fig. 2: CCDF of $\rho(\mathbf{S})$, signal-driven sets ($N_{\text{sc}} = 256$, $\eta = 0.8$).

[44] L. Cho and S.-W. Chang, "On the performance of discrete Papoulis–Gerchberg type iterative reconstruction," *IEEE Signal Process. Lett.*, 2026.

NMSE Decomposition

Active-subcarrier NMSE after $i_{\max}$ iterations:

$$\varepsilon(\eta, \gamma_s, i_{\max}) = \frac{\mathbb{E}[\|\hat{\mathbf{s}} - \mathbf{s}\|^2]}{\mathbb{E}[\|\mathbf{s}\|^2]}$$

Given the partition, the receiver chain is linear: $\hat{\mathbf{s}} = \hat{\mathbf{s}}_0 + \mathbf{A}\mathbf{w}$, cross term = 0, $\sigma_w^2 = L/\gamma_s$

$$\varepsilon(\eta, \gamma_s, i_{\max}) = \Psi(\eta, i_{\max}) + \frac{\Xi(\eta, i_{\max})}{\gamma_s}$$

- ▶ Ψ : SNR-independent **signal-residual floor**
- ▶ Ξ : γ_s -independent **noise coefficient**
- ▶ exact second-moment identity, coefficients from profiling once per $(R, L, i_{\max})$
- ▶ near-threshold misclassification coupling: weak, absorbed, no separate term needed

Two Terms, Opposite Pulls

$\Psi(\eta)$: monotone decreasing

- ▶ init at received magnitude $\mathcal{C}(r) \Rightarrow$ initial residual = compression gap $r - \mathcal{C}(r) \rightarrow 0$ as $\eta \rightarrow 1$
- ▶ larger η : fewer, smaller companded peaks
- ▶ at $\eta = 1$: PGIR off, Ψ jumps to the hard-clip floor of [37] (star marker)

$\Xi(\eta)$: U-shaped

- ▶ small η : big unknown set, more amplified noise
- ▶ $\eta \rightarrow 1^-$: thin band $[A_{\text{th}}, R]$, peaks crowd the threshold, noise-sensitive reconstruction
- ▶ shallow valley, minimum near $\eta \approx 0.91$ ($i_{\text{max}} \geq 5$)

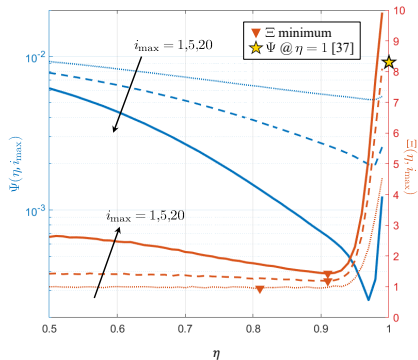


Fig. 3: Ψ, Ξ for $i_{\text{max}} \in \{1, 5, 20\}$ ($R = 4$ dB, $L = 4$).

[37] M. Hu et al., *IEEE Trans. Broadcast.*, 2025.

Three-Regime Trajectory of η_{opt}

$$\eta_{\text{opt}}(\gamma_s, i_{\text{max}}) = \arg \min_{\eta \in (0,1]} \left[\Psi + \frac{\Xi}{\gamma_s} \right], \quad \gamma^* \triangleq \inf \{ \gamma_s : \eta_{\text{opt}} < 1 \} \text{ (engagement onset)}$$

Regime I, low SNR. Ξ/γ_s dominates, reconstruction = net noise loss $\Rightarrow \eta_{\text{opt}} = 1$, framework = [37].
Structural consequence of the decomposition

Regime II, the dip. Hard-clip floor overtaken $\Rightarrow \eta_{\text{opt}}$ drops to the Ξ -minimum, ≈ 0.80 – 0.91 across R, L . Onset 19–24 dB E_s/N_0

Regime III, high SNR. $\varepsilon \rightarrow \Psi(\eta) \Rightarrow$ climb back toward 1^- , shrinking the discarded set and the residual floor

Non-monotone trajectory: $1 \rightarrow \text{dip} \rightarrow 1^-$. What i_{max} buys: NMSE depth at the dip, not its location.

Trajectories: AWGN vs. SR-AS

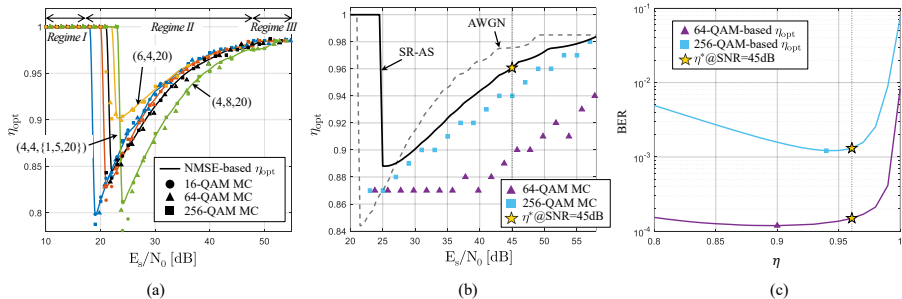


Fig. 4: (a) AWGN, labels $(R$ [dB], L , i_{max}). (b) SR-AS ($R = 4$ dB, $L = 4$, $i_{\text{max}} = 20$). (c) BER vs. η at $\gamma_s = 45$ dB.

- dip deepens with i_{max} , shallower at larger R , slightly deeper at larger L
- SR-AS: average the AWGN NMSE over the fading law, **no re-simulation**. Deep-fade tail $\Rightarrow$ onset shifts up 3–6 dB, delayed return to unity
- MC BER optima (markers) track the NMSE rule

NMSE as a BER Proxy

Lemma 1: residual-power monotonicity

$\mathbf{e} = \sqrt{\Psi}\boldsymbol{\xi}$, $\boldsymbol{\xi}$ η -invariant $\Rightarrow$ SER non-decreasing in Ψ (convex Voronoi cells). Gray mapping $\Rightarrow$ high-SNR BER inherits it.

Proposition 1: modulation-free selector

BER strictly increasing in $\varepsilon = \Psi + \Xi/\gamma$. Noise-dominated: $\text{BER} \approx f_{\mathcal{M}}(1/\varepsilon)$ [49]. Either regime:

$$\eta_{\text{opt}}^{\text{BER}} = \arg \min_{\eta} \text{BER} = \arg \min_{\eta} \varepsilon = \eta_{\text{opt}}^{\text{NMSE}}, \quad \mathcal{M} \text{ enters the map only}$$

- ▶ AWGN check: 64/256-QAM MC optima agree with NMSE rule to **within 0.02** in η
- ▶ high SNR: BER below Monte-Carlo resolution, NMSE still resolvable $\Rightarrow \eta_{\text{opt}}$ at **no sampling cost**

[49] K. Cho and D. Yoon, "On the general BER expression of one- and two-dimensional amplitude modulations," *IEEE Trans. Commun.*, 2002.

Fading Bias of the BER Optimum

Proposition 2

Flat fading, perfect CSI, errors accrue where $B_{\mathcal{M}}$ is convex, $\varepsilon(\eta^*, \gamma_s | h|^2) \leq 1/(\mathcal{M} - 1)$:

$$\left. \frac{\partial \overline{\text{BER}}_{\mathcal{M}}}{\partial \eta} \right|_{\eta=\eta^*} \geq 0 \quad \Rightarrow \quad \eta_{\text{opt}}^{\text{BER}} \leq \eta^*$$

Displacement = deep-fade weighting / valley curvature (Chebyshev association inequality).

At $\gamma_s = 45$ dB [Fig. 4(c)]:

- ▶ **256-QAM**: sharp valley, $\eta^* = 0.96$ vs. $\eta_{\text{opt}}^{\text{BER}} = 0.94$, penalty 0.34 dB
- ▶ **64-QAM**: flat valley, η^* on the shoulder, penalty 0.99 dB
- ▶ one ε -selected η serves all orders: < 0.8 dB (256-QAM), < 1.8 dB (64-QAM) across the resolvable range.
Accurate where it matters, harmless where it does not

CSI-Gated Engagement

Per-symbol AWGN at $\gamma_s|h|^2 \Rightarrow$ engage/bypass by a single crossover $\gamma_g(\mathcal{M})$:

$$B_r(\gamma_g) = B_c(\gamma_g), \quad \text{mode}(h) = \begin{cases} \text{reconstruct,} & \gamma_s|h|^2 > \gamma_g, \\ \text{bypass,} & \text{otherwise} \end{cases}$$

Proposition 3: CSI-gate optimality

$$\overline{\text{BER}}_{\mathcal{M}}^{\text{gate}}(\gamma_s) \leq \overline{\text{BER}}_{\mathcal{M}}^{\text{bypass}}(\gamma_s) \quad \forall \mathcal{M}, \forall \gamma_s$$

Pointwise minimum per symbol, preserved by the fading expectation.

- ▶ **provably never worse than the hard-clip baseline**, still harvests the shallow-fade gain
- ▶ γ_g calibrated offline once per constellation, from the same profiles that supply η_{opt}
- ▶ transmit CSI mild on the quasi-static feeder link: TDD reciprocity or NTN CSI reports [16]

[16] 3GPP, "Solutions for NR to support non-terrestrial networks (NTN) (Release 16)," TR 38.821 V16.2.0, 2023.

Simulation Setup

- ▶ baseline: hard limiter [37] = $\eta = 1$ corner of the framework
- ▶ PAPR / spectrum: 16-QAM (envelope metrics order-insensitive)
- ▶ detection: 16- / 64- / 256-QAM
- ▶ $\eta \in \{0.7, 0.8, 0.9\}$ vs. $\eta = 1$ vs. un-companded, all energy-conserving $\Rightarrow$ **equal average power**
- ▶ $\text{IBO} = R + 1$ dB: 1-dB peak back-off rule

Table I: system and simulation parameters.

Parameter	Value
Active subcarriers N_{sc}	256
Oversampling factor L	4, 8 ($N = 1024, 2048$)
Modulation order	16-, 64-, 256-QAM
PAPR amplitude target R	$\{4, 6\}$ dB
PGIR iterations i_{max}	5 (swept $\{1, 5, 20\}$)
Input back-off (IBO)	$R + 1$ dB
SSPA (Rapp) sharpness p_{rapp}	3
TWTA AM/AM (α_S, β_S)	(2.1587, 1.1517)
TWTA AM/PM (α_Φ, β_Φ)	(4.0033, 9.1040)
SR-AS channel (b_0, m, Ω)	(0.126, 10.1, 0.835)
Rician factor K_{ric}	3.31 (5.2 dB)

[37] M. Hu et al., *IEEE Trans. Broadcast.*, 2025. [46] C. Rapp, 1991. [47] A. A. M. Saleh, *IEEE Trans. Commun.*, 1981. [48] A. Abdi et al., *IEEE Trans. Wireless Commun.*, 2003.

PAPR: Peak Reduction Costs Nothing

- ▶ un-companded tail: ≈ 12 dB at 10^{-4}
- ▶ all variants: near-deterministic ceiling at R , reductions ≈ 8 dB ($R = 4$) / 6 dB ($R = 6$)
- ▶ smaller η : peak only a few tenths of a dB below the hard limiter

Takeaway

Smooth companders match [37] in peak reduction
 $\Rightarrow$ spectral + reconstruction gains come **free in PAPR**.

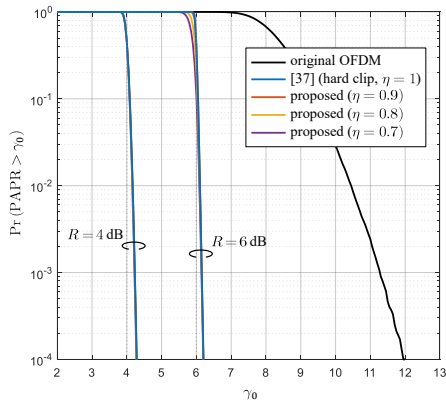


Fig. 5: PAPR CCDF, $R \in \{4, 6\}$ dB ($N_{\text{sc}} = 256$, $L = 4$, 16-QAM).

[37] M. Hu et al., *IEEE Trans. Broadcast.*, 2025.

Spectral Containment

- ▶ close-in shoulder: mildly higher as $\eta \downarrow$ (wider $[\eta R, R]$), lower at $R = 6$ dB
- ▶ far-out: ordering **reverses**. $\mathcal{O}(1/f^3)$ (C^1) vs. $\mathcal{O}(1/f^2)$ (C^0): **20 dB/decade steeper**
- ▶ crossover near $2/3$ of the sampling rate, beyond it proposed $< [37]$
- ▶ far-out tail governs spectral-mask and adjacent-channel compliance

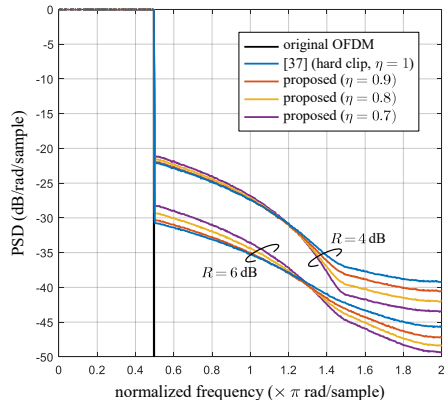


Fig. 6: PSD, normalized to in-band level ($N_{sc} = 256$, $L = 4$).

[37] M. Hu et al., *IEEE Trans. Broadcast.*, 2025.

Convergence and Oversampling

- ▶ AWGN, 16-QAM, $R = 4$ dB, ideal front end: reconstruction isolated
- ▶ below γ^* : ride the baseline. Above: **separate downward**, no flattening over the plotted range
- ▶ most gain in the first few iterations $\Rightarrow i_{\max} = 5$
- ▶ larger L **hurts**: per-sample SNR $= \gamma_s/L$, -3 dB per doubling on the sub-threshold samples PGIR extrapolates from. Finite-iteration *noise* effect, not conditioning
- ▶ $L \geq 4$ needed for PAPR-faithful peak capture $\Rightarrow$ **optimum $L = 4$**

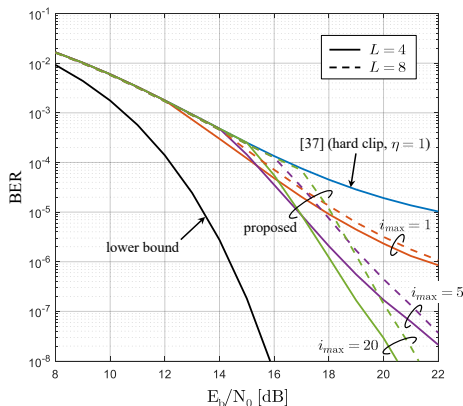


Fig. 7: BER in AWGN, $L \in \{4, 8\}$, $i_{\max} \in \{1, 5, 20\}$.

Robustness Across Amplifiers

- ▶ SR-AS, 64-QAM, $R = 6$ dB, $i_{\max} = 5$
- ▶ separation **widens with amplifier severity**:
 - ▷ **ideal PA**: both near the bound
 - ▷ **Rapp SSPA**: shallow baseline floor, proposed stays on the bound
 - ▷ **Saleh TWTA**: baseline hits an error floor, proposed **descends past it**, zero amplifier knowledge
- ▶ largest benefit exactly in the TWTA regime of high-power transponders
- ▶ CSI gate keeps proposed $\leq$ baseline in every regime

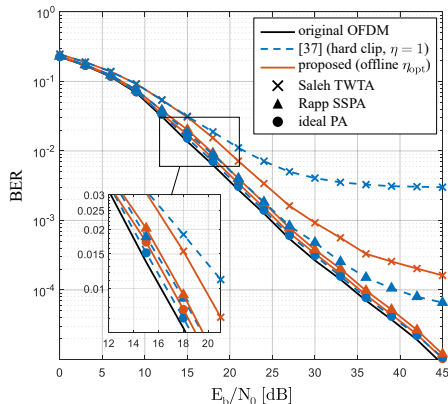


Fig. 8: BER over SR-AS, 64-QAM, three amplifier regimes.

[37] M. Hu et al., *IEEE Trans. Broadcast.*, 2025.

[46] C. Rapp, 1991.

[47] A. A. M. Saleh, *IEEE Trans. Commun.*, 1981.

Modulation-Order Scaling: the Central Result

- ▶ Saleh TWTA. Gain over [37]: negligible at 16-QAM, clear at 64-QAM, **decisive at 256-QAM**
- ▶ error geometry: reconstruction improves the shallow-fade **bulk**, injects error in deep fades. Tighter constellations err in the bulk $\Rightarrow$ gain grows with order
- ▶ AM/PM sets the 256-QAM plateau, still far below the baseline floor
- ▶ CSI gate: 16-QAM transient $\rightarrow$ uniform gain, 64/256-QAM gated $\equiv$ offline

The high-order links that motivate the work are exactly where reconstruction gains most.

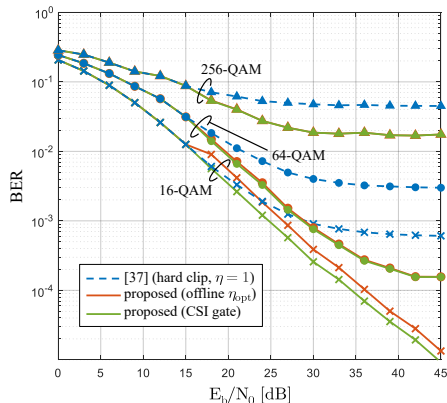


Fig. 9: BER over SR-AS, Saleh TWTA, 16- / 64- / 256-QAM.

[37] M. Hu *et al.*, "Near-optimal piecewise linear companding transform for PAPR reduction of OFDM systems," *IEEE Trans. Broadcast.*, 2025.

Computational Complexity

- ▶ **Transmitter** $\equiv$ [37]: per-sample memoryless map before the IFFT. No transform, no signaling, no bandwidth expansion
- ▶ all added cost at the **resource-rich gateway**: $2i_{\max}$ FFT/IFFT per engaged symbol, $\mathcal{O}(i_{\max} N \log N)$. At $i_{\max} = 5$: **10 length- N transforms**
- ▶ FFT = the OFDM modem's native primitive, dedicated hardware simply reused
- ▶ CSI gate runs the loop only where reconstruction pays $\Rightarrow$ lower average overhead

Versus receiver-side alternatives

Per iteration $\approx$ decision-aided regeneration [38]. CS receiver [40]: costlier sparse solver. Equalizer [41]: explicit PA model required. PGIR: pure transform recursion, no decisions, no priors, no models.

[38] H. Chen and A. M. Haimovich, *IEEE Commun. Lett.*, 2003. [40] E. B. Al-Safadi and T. Y. Al-Naffouri, *IEEE Trans. Signal Process.*, 2012. [41] J. He et al., *IEEE Trans. Commun.*, 2024.

Conclusion

- ▶ **Discard and reconstruct:** C^1 compander co-designed with gateway-side PGIR, transparent, no side information, no amplifier model
- ▶ PAPR capped, $\mathcal{O}(1/f^3)$ OOB decay, peaks rebuilt from the null-subcarrier subspace
- ▶ two-term NMSE decomposition $\Rightarrow$ **three-regime** $\eta_{\text{opt}}(\gamma_s)$, hard clipping [37] as the principled $\eta = 1$ limit
- ▶ BER = monotone in NMSE, order-independent $\Rightarrow \eta$ from **cheap profiling**, resolvable beyond Monte-Carlo reach
- ▶ gains widen with amplifier severity and modulation order, **decisive under the TWTA**, CSI gate provably $\leq$ baseline in deep fades

Future work

Coded performance, imperfect CSI + receiver-side gate variant, other peak-limited waveforms.

Appendix

A: C^1 tail rate and uniqueness of g

B: noise-unbiased AGC scalar

Appendix A: C^1 Tail Rate and Uniqueness of g

C^0 **for free:** both branches meet at the threshold, $\mathcal{C}(A_{\text{th}}^-) = gA_{\text{th}} = \eta R = \mathcal{C}(A_{\text{th}}^+)$, any β

$$C'(r) = \begin{cases} g, & 0 \leq r < A_{\text{th}}, \\ 2\beta(1 - \eta)Rr e^{-\beta(r^2 - A_{\text{th}}^2)}, & r > A_{\text{th}} \end{cases}$$

C^1 **matching:** $C'(A_{\text{th}}^+) = 2\beta(1 - \eta)RA_{\text{th}} \stackrel{!}{=} g$, with $A_{\text{th}} = \eta R/g$

$$\beta = \frac{g}{2(1 - \eta)RA_{\text{th}}} = \frac{g^2}{2\eta(1 - \eta)R^2} \quad \text{positive, finite on all of } \eta \in (0, 1)$$

- ▶ $\eta \rightarrow 1^-$: $\beta \rightarrow \infty$, degenerates to the C^0 hard limiter
- ▶ **uniqueness of g :** both branches rise pointwise with g , threshold $A_{\text{th}} = \eta R/g$ falls $\Rightarrow \mathbb{E}[|\mathcal{C}(r)|^2]$ continuous, strictly increasing in g
- ▶ range: 0 (as $g \rightarrow 0^+$) $\rightarrow R^2 > 1$ (as $g \rightarrow \infty$) $\Rightarrow$ **exactly one root** of the energy equation for every $\eta \in (0, 1)$

Appendix B: Noise-Unbiased AGC Scalar

One complex scalar refers $\mathbf{u} = \mathcal{G}_{\text{pa}}(\tilde{\mathbf{x}})$ back to the transmit-envelope domain, compensating *average* gain compression and phase rotation only:

$$G_{\text{agc}} = \sqrt{\frac{\mathbb{E}[|\tilde{x}|^2]}{\mathbb{E}[|u|^2]}} e^{-j\angle \mathbb{E}[u\tilde{x}^*]}$$

- ▶ both moments = received power + received-reference correlation, already estimated from pilot / decision-directed averages $\Rightarrow$ **no PA knowledge**
- ▶ zero-mean noise: no contribution to $\mathbb{E}[u\tilde{x}^*]$, only an η -independent constant in $\mathbb{E}[|u|^2]$, removable by a noise-power estimate $\Rightarrow$ asymptotically unbiased
- ▶ backed-off SSPA: $\angle \mathbb{E}[u\tilde{x}^*] \approx 0$, $|G_{\text{agc}}| \approx 1$, correction omitted
- ▶ TWTA: phase term **essential**, uncompensated AM/PM would dominate the high-order QAM error
- ▶ detection threshold held at ηR after referral, soft-knee mismatch absorbed into the two-term decomposition